

MedTrackPNG

Version 2.0

Integrated Health Centre, Clinical & Pharmacy Management System

PROJECT DOCUMENTATION

Project	MedTrackPNG
Version	2.0
Document type	Project Documentation
Status	Active Development
Approach	Offline-first Android application
Primary environment	Papua New Guinea health facilities
Co-creators	Ends and Francis Lui
Year	2026

This document describes intended project architecture, functions, responsibilities and development direction. Individual system functions remain subject to implementation, testing, validation and future revision.

1. Project Overview

MedTrackPNG is a proposed offline-first digital health-centre management application designed for use by health facilities in Papua New Guinea.

The system is intended to provide a single facility-level platform for managing:

- Patient registration and records
- Clinical encounters
- Prescriptions
- Medicine catalogues
- Pharmacy dispensing
- Medicine batches and expiry dates
- Stock transactions
- Requisitions and receiving
- Reports
- User accounts and role-based access
- Audit trails
- Future data synchronisation and integration

The application is being designed specifically with the operational realities of PNG health facilities in mind, including unreliable internet connectivity and the need for accurate medicine and patient records.

2. Project Name and Identity

Item	Detail
Project name	MedTrackPNG
Current development version	MedTrackPNG v2.0
Project type	Health Facility Management / Clinical Information / Pharmacy & Inventory Management System
Development approach	Offline-first Android application with future synchronisation and integration capability

3. Project Creators and Co-Authors

3.1 Creator / Co-Author — Ends

Role: Co-Creator / Co-Author

Project responsibility:

- Overall project concept and direction
- Application requirements
- System design and functionality
- Health-facility workflow requirements
- Project coordination
- Application development direction
- User-interface and operational requirements

- Overall project documentation

3.2 Co-Creator / Co-Author — Francis Lui

Field	Detail
Date of birth	4 July 1998
Job title	Pharmacy Assistant
Place of origin	Manus
Citizenship	Papua New Guinea
Workplace	Kimadan Health Center
Position	PS06 NIPHA
Department	Kavieng Provincial Pharmacy

Francis Lui contributes specialist pharmacy and medicine-management knowledge to the project, including:

- Pharmacy workflow requirements
- Medicine dispensing procedures
- Medicine catalogue requirements
- Stock management requirements
- Batch and expiry management
- Pharmacy requisition and receiving processes
- Medicine availability and inventory requirements
- Clinical/pharmacy workflow considerations
- Practical health-facility pharmacy requirements
- Review of pharmacy-related system functionality

His practical pharmacy experience is intended to help ensure that MedTrackPNG reflects real operational requirements rather than being designed solely from a software-development perspective.

4. Project Purpose

The purpose of MedTrackPNG is to develop a practical digital platform that assists health-centre staff in managing patient, clinical, pharmacy and inventory information in one integrated system.

The system is intended to reduce dependence on disconnected paper records and separate manual processes while maintaining an operational model suitable for facilities where internet connectivity may be unavailable or unreliable.

5. Project Objectives

The primary objectives are to:

1. Develop an offline-capable health-centre management application.
2. Improve the organisation and accessibility of patient records.
3. Support structured clinical encounter documentation.
4. Connect prescriptions with pharmacy dispensing.
5. Maintain accurate medicine and stock records.

6. Track medicine batches and expiry dates.
7. Support FEFO — First Expiry, First Out — stock management.
8. Record every significant stock movement as a transaction.
9. Provide role-based access to different categories of staff.
10. Maintain an audit trail of important system activities.
11. Reduce duplication and manual errors.
12. Provide operational reports for authorised users.
13. Establish a foundation for future online synchronisation.
14. Provide a future integration boundary for external systems and national supply-chain platforms where appropriate.

6. Core System Philosophy

MedTrackPNG v2.0 is based on several fundamental principles.

6.1 Offline First

The application must remain useful when internet connectivity is unavailable. Core facility operations should therefore be stored locally on the device and should not depend on continuous internet access.

6.2 Data Integrity

Important information should not simply be overwritten. Medicine stock should be represented through transactions such as:

- Opening balance
- Receipt
- Dispensing
- Adjustment
- Damage/loss
- Transfer
- Stock correction

This creates a traceable stock history.

6.3 Role-Based Access

Different staff members should have different access according to their responsibilities. A user should not receive unrestricted access simply because they can log into the application.

6.4 Auditability

Important actions should be recorded, including:

- User
- Action
- Date/time
- Record affected
- Relevant transaction

6.5 Future Integration

The application should be developed with a clean synchronisation/API boundary so that future integration with other health information or supply-chain systems does not require rebuilding the application from the beginning.

7. Proposed User Roles

Role	Access and functions
Administrator	Full system administration: user management; role and permission management; system settings; reports; audit logs; data-management functions.
Assistant Pharmacist / Dispenser	Primary pharmacy functions: medicine catalogue; stock; batch management; expiry monitoring; dispensing; receiving; requisitions; pharmacy reports.
Clinical Staff	Patient registration; patient records; clinical encounters; prescriptions; appropriate clinical information.
Nurse	Access to patient and clinical functions according to assigned permissions.
Other Staff	Access limited to functions specifically authorised by the administrator.

8. Core Modules

8.1 Authentication and Security

The system will provide:

- User login
- Session management
- Role identification
- Permission enforcement
- Secure credential handling
- Sign-out
- Audit logging

The login architecture is specifically designed to prevent the previous problem where the login screen remained visible after successful authentication.

8.2 Dashboard

The dashboard will provide authorised users with access to the functions available to their role. Potential information includes:

- Current stock alerts
- Expiring medicines
- Low-stock medicines
- Recent dispensing activity
- Patient activity
- Pending requisitions
- Receiving activity
- System notifications

8.3 Patient Management

The patient module will support:

- Patient registration
- Patient identification
- Demographic information
- Contact information where appropriate
- Patient history
- Encounter history
- Prescription history
- Pharmacy/dispensing history where authorised

8.4 Clinical Encounters

The clinical module will support structured recording of:

- Patient visits
- Clinical assessments
- Diagnoses
- Treatment
- Clinical notes
- Prescriptions
- Follow-up information

Access will be controlled according to staff role.

9. Pharmacy and Medicine Management

9.1 Medicine Catalogue

The catalogue will provide structured medicine information, including appropriate fields for:

- Medicine name
- Generic name
- Strength
- Dosage form
- Unit
- Catalogue/reference information
- Active/inactive status

9.2 Batch Management

Medicines will be tracked by batch where applicable. A batch record can contain:

- Medicine
- Batch number
- Quantity
- Expiry date
- Receiving information
- Stock location
- Status

9.3 Expiry Management

The system will identify medicines approaching expiry and allow authorised pharmacy personnel to monitor them. Expiry monitoring will support appropriate stock-management decisions.

9.4 FEFO

MedTrackPNG will support First Expiry, First Out (FEFO). Where multiple batches of the same medicine are available, the system can identify the batch with the earliest appropriate expiry date for dispensing.

10. Stock Ledger

Stock will be transaction-based. Instead of treating stock as a number that can simply be overwritten, every significant movement will create a transaction.

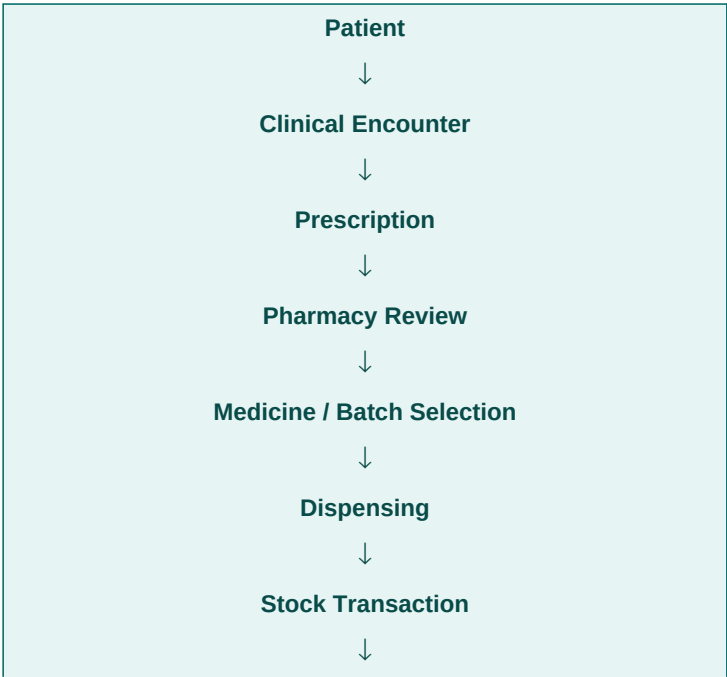
Transaction	Quantity
Opening Balance	+1,000
Received	+500
Dispensed	-120
Damaged	-5
Adjustment	-10
Current Balance	1,365

Illustrative stock ledger example.

Each transaction should be associated with the relevant medicine and, where applicable, batch. This approach provides a historical record of how the current stock balance was produced.

11. Prescription and Dispensing Workflow

The intended workflow is:



Audit Record

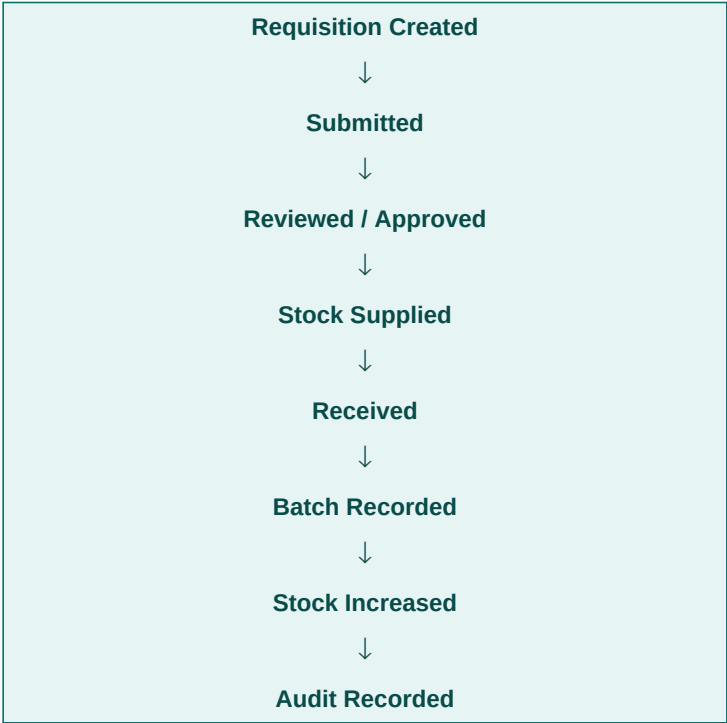
End-to-end clinical-to-pharmacy information chain.

This creates a direct relationship between the clinical prescription, the medicine actually dispensed and the resulting stock movement.

12. Requisitions and Receiving

The system will support facility-level stock requests and receiving processes.

Potential workflow:



Traceable incoming-stock workflow.

This creates a traceable history for incoming stock.

13. Audit Trail

Important actions will generate audit records. An audit record may contain:

- User
- Role
- Action
- Date/time
- Record identifier
- Previous value where appropriate
- New value where appropriate
- Transaction reference

Examples include:

- User login
- User creation

- Patient creation
- Prescription creation
- Medicine dispensing
- Stock adjustment
- Batch creation
- Requisition approval
- Receiving
- User permission changes

14. Database Architecture

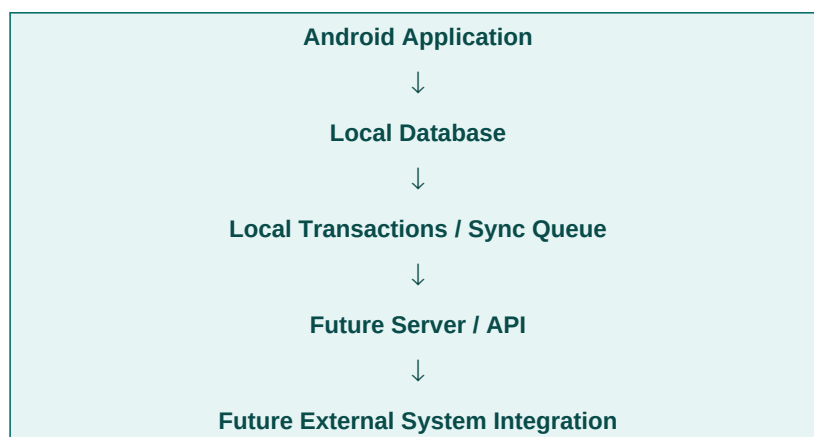
The planned local database will include core entities such as:

- User
- Role
- Permission
- Patient
- Encounter
- Prescription
- Prescription Item
- Medicine
- Batch
- Stock Transaction
- Dispensing Record
- Requisition
- Requisition Item
- Audit Log
- Sync Record

The database will be designed around relationships rather than disconnected screens.

15. Offline and Synchronisation Architecture

MedTrackPNG will be designed as an offline-first system. The basic architecture is:



Offline-first architecture with a future sync boundary.

This allows the application to continue operating locally while preparing information for later synchronisation. Synchronisation should use transaction/event-based records rather than blindly overwriting entire datasets.

16. Security

Security requirements include:

- Role-based access control
- Secure password handling
- Session management
- Restricted administrative functions
- Audit logging
- Local data protection
- Controlled data modification
- Permission enforcement at the application/data layer

Development credentials must not be used in a production release.

17. Development Roadmap

Phase	Focus	Key work
1 — Foundation	Platform	Android project; application architecture; database foundation; navigation; authentication.
2 — Users & Roles	Access control	Users; roles; permissions; session management; audit foundation.
3 — Patient & Clinical	Clinical care	Patient registration; patient records; clinical encounters; prescriptions.
4 — Pharmacy	Dispensing	Medicine catalogue; pharmacy workflow; dispensing; batch management; expiry management.
5 — Inventory	Stock control	Stock ledger; receiving; requisitions; adjustments; transfers; FEFO.
6 — Reporting	Visibility	Stock, expiry, dispensing, patient/clinical and operational reports.
7 — Synchronisation	Connectivity	Sync queue; conflict handling; API layer; future external-system integration.
8 — Testing	Quality	Authentication, permissions, records, workflows, stock, batches, expiry, audit, offline, sync, navigation, persistence.
9 — Release	Deployment	Debug testing; release build; APK installation; device testing; UAT; production security review; release documentation.

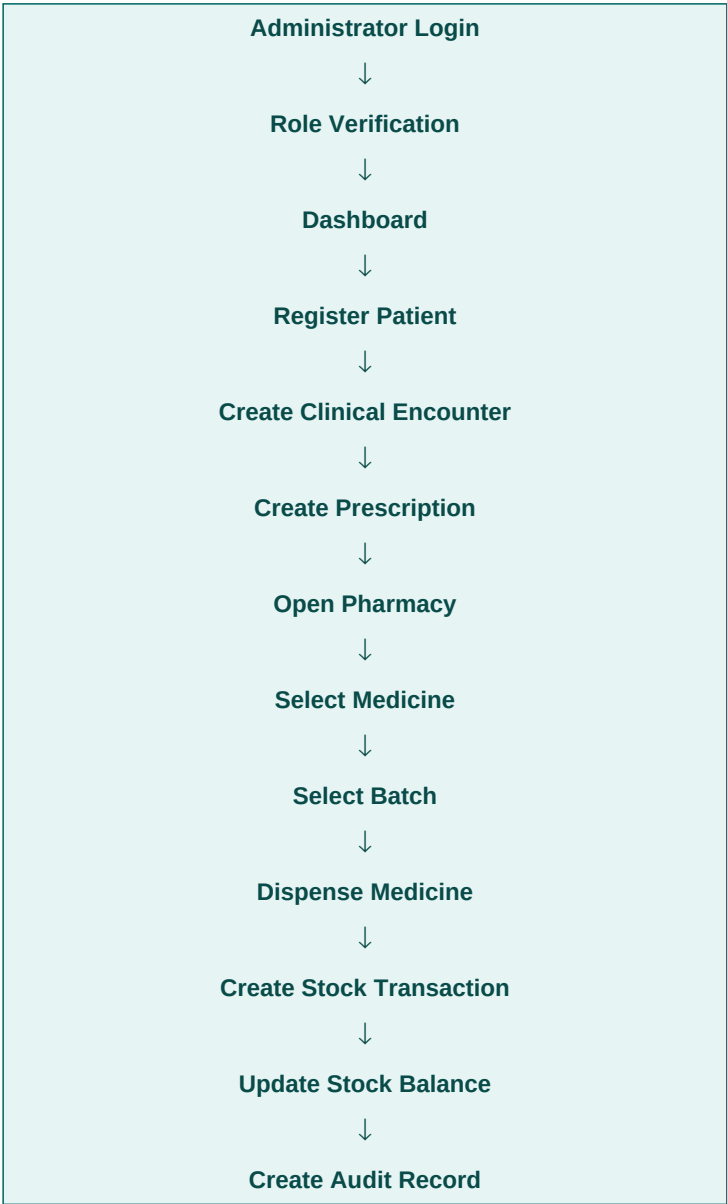
Phase 8 testing will cover:

- Authentication
- Permissions
- Patient records
- Clinical workflow
- Prescriptions
- Dispensing
- Stock calculations
- Batch tracking

- Expiry handling
- Audit records
- Offline operation
- Synchronisation
- Navigation
- Data persistence

18. First End-to-End Functional Test

The first major test of MedTrackPNG v2.0 will be the following chain. The entire chain must work before the project is considered to have achieved its first functional milestone.



First functional milestone — end-to-end test chain.

19. Project Responsibilities

Co-author	Primary responsibilities
Ends	Overall project direction; software/application development; system architecture; application design; user interface; technical implementation; documentation; testing coordination; future system integration.
Francis Lui	Pharmacy workflow expertise; medicine-management requirements; dispensing workflow; stock-management requirements; batch and expiry requirements; requisition and receiving requirements; pharmacy process review; practical validation of pharmacy functionality.

The project therefore combines software development and practical pharmacy expertise.

20. Intended Outcome

The intended outcome is a practical, maintainable and expandable digital health-centre platform that can support day-to-day facility operations while remaining usable in environments where connectivity may be limited.

MedTrackPNG v2.0 is not intended to be merely a collection of digital forms. The objective is to create an integrated system in which:

Patient information → Clinical care → Prescription → Pharmacy → Dispensing → Stock → Audit

form one connected information chain.

21. Project Authorship

Co-Creators / Co-Authors

#	Name	Attribution
1	Ends	Project Creator / Co-Author
2	Francis Lui	Pharmacy Assistant, Kimadan Health Center, PS06 NIPHA, Kavieng Provincial Pharmacy

Document Status

Field	Value
Project	MedTrackPNG
Version	2.0
Document type	Project Documentation
Status	Active Development
Development approach	Offline-first Android application
Primary environment	PNG Health Facility
Co-creators	Ends and Francis Lui

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22.2 Co-Creators / Co-Authors

#	Name	Role and attribution
1	Ends	Project Creator / Co-Author. Overall concept, system design, application development direction, architecture, documentation and technical coordination.
2	Francis Lui	Co-Creator / Co-Author. Pharmacy Assistant, Kimadan Health Center, PS06 NIPHA, Kavieng Provincial Pharmacy. Specialist contribution on pharmacy workflow, medicine management, dispensing, batch and expiry, stock, requisitions and practical facility requirements.

22.3 Short Copyright Notice (for app, APK, README or About screen)

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Co-creators: Ends and Francis Lui.

MedTrackPNG is an offline-first health-centre, clinical and pharmacy management system designed for Papua New Guinea health facilities.

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Created by Ends and Francis Lui.

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